MINI PROJECT REPORT

Enterprise Student Record and Academic Management Platform

Unit 1: Design Thinking and Product Planning

Prepared for College Submission

Student Name: waghmare suyash atish
Roll No.: _2126UMLM2533
Class / Division:DSY AIML
Academic Year: 2026–27

CERTIFICATE

This is to certify that the mini project titled “Enterprise Student Record and Academic Management Platform” has been prepared as part of Unit 1 project work. The project applies Design Thinking and product planning activities to a student academic management problem.

Guide/Faculty Signature: _____

Date: _____

INDEX

1. 1. Introduction

2. 2. Business Understanding

3. 3. Problem Statement

4. 4. Design Thinking – Empathize

5. 5. Design Thinking – Define

6. 6. Design Thinking – Ideate

7. 7. Design Thinking – Prototype

8. 8. Design Thinking – Test

9. 9. IBM Design Thinking

10. 10. IBM Loop

11. 11. Stakeholder Analysis

12. 12. User Personas

13. 13. Functional Requirements Specification (FRS)

14. 14. Non-Functional Requirements Specification (NFRS)

15. 15. Product Vision

16. 16. Business Objectives

17. 17. Minimum Viable Product (MVP)

18. 18. Product Roadmap

19. 19. Initial System Architecture

20. 20. Project Repository Structure

21. 21. Project Charter

22. 22. Conclusion

1. Introduction

The Enterprise Student Record and Academic Management Platform is a proposed software product for managing student academic information in a higher education institution. The system is designed to reduce problems caused by manual record management and provide a centralized platform for student registration, course information, attendance, examinations, results, and academic reports.

The project follows a human-centred Design Thinking approach. Instead of starting directly with coding, the project first studies users, their problems, stakeholders, requirements, and the desired product outcomes.

2. Business Understanding

The main academic processes covered by the platform are student admission/registration, course registration, attendance management, examination management, result management, and report generation.

Manual management can cause duplicate records, slow searching, calculation mistakes, difficulty in preparing reports, and communication delays. The proposed platform provides centralized records and faster access to academic information.

3. Problem Statement

Students and academic staff may face difficulty when academic records are maintained through separate or manual processes. Students need quick access to their academic information, while faculty and administrators need accurate records and reports. The project therefore proposes a centralized platform to make academic record management simpler, faster, and more reliable.

4. Design Thinking – Empathize

The first stage is to understand users and their problems. The main user groups are students, faculty members, academic administrators, examination cell staff, and IT support personnel.

Students need easy access to profiles, attendance, results, and reports. Faculty members need efficient attendance and academic record management. Administrators need centralized information and reports. Examination staff need accurate examination and result records.

5. Design Thinking – Define

Core problem statement: “Students and academic staff need a simple and centralized system for managing academic records because manual or disconnected processes can make registration, attendance, examination, result management, and reporting slow and error-prone.”

6. Design Thinking – Ideate

Possible solution ideas include a centralized student database, role-based login, online student registration, attendance entry, examination and result modules, automatic academic reports,

dashboards, and searchable student records. The first release should focus on essential academic-management functions.

7. Design Thinking – Prototype

A low-fidelity prototype can contain the Login Page, Student Dashboard, Faculty Dashboard, Administrator Dashboard, Student Registration Form, Attendance Page, Examination/Result Page, and Report Page. The prototype should show navigation and basic information flow before detailed development.

8. Design Thinking – Test

The prototype can be shown to representative students, faculty members, and administrators. Feedback can be collected on ease of navigation, clarity of forms, required information, and task completion. Changes should be made based on user feedback before development.

9. IBM Design Thinking

The project follows three IBM Design Thinking principles from the guide:

1. Focus on User Outcomes – measure whether the platform makes academic tasks easier for users.

2. Restless Reinvention – continuously improve the product using feedback and observations.

3. Diverse Empowered Teams – involve different roles such as product, UX, development, QA, database, and operations stakeholders.

10. IBM Loop

The IBM Loop is Observe → Reflect → Make.

Observe: study user interactions, problems, and feedback.

Reflect: analyze observations and decide what should be improved.

Make: design or modify the solution and test the improvement.

The cycle is repeated for continuous product improvement.

11. Stakeholder Analysis

The key stakeholders are students, faculty members, academic administrators, examination cell staff, and IT support personnel. Students use the platform for academic information. Faculty members manage academic activities. Administrators monitor records and reports. Examination staff manage examination and result information. IT support maintains the technical environment.

12. User Personas

Persona 1 – Student: Goal: view profile, attendance, examination and results quickly. Challenges: difficulty obtaining information from different sources. Expectation: simple dashboard and accurate records.

Persona 2 – Faculty Member: Goal: manage attendance and academic information efficiently. Challenges: repeated manual data entry and report preparation. Expectation: easy forms and reliable records.

Persona 3 – Administrator: Goal: monitor academic records and generate reports. Challenges: lack of centralized information. Expectation: dashboard, search, and reporting functions.

13. Functional Requirements Specification (FRS)

FR-01: User authentication and role-based access.

FR-02: Student registration and profile management.

FR-03: Course registration and course information management.

FR-04: Attendance recording and viewing.

FR-05: Examination information management.

FR-06: Result entry and result viewing.

FR-07: Academic report generation.

FR-08: Search and view student records.

FR-09: Administrator management of academic data.

14. Non-Functional Requirements Specification (NFRS)

NFR-01 Security: user data should be protected and access should be role-based.

NFR-02 Usability: screens and forms should be simple to understand.

NFR-03 Reliability: records should be stored accurately and consistently.

NFR-04 Performance: common searches and pages should respond quickly.

NFR-05 Scalability: the platform should be able to support increasing numbers of students and records.

NFR-06 Maintainability: the system should have a clear and organized structure.

15. Product Vision

Vision: “To provide a simple, centralized, and reliable digital platform that helps higher-education institutions manage student academic records and enables students and staff to access important academic information efficiently.”

Mission: Reduce manual academic-record work, improve accessibility of information, and support accurate academic management.

16. Business Objectives

1. Centralize student academic information.

2. Reduce manual record-management work.

3. Make attendance and result information easier to access.

4. Improve accuracy and consistency of academic records.

5. Support faster report generation.

6. Provide a foundation for future Agile development.

17. Minimum Viable Product (MVP)

The first release should contain:

1. Student registration and profile.

2. Basic student record management.

3. Course information.

4. Attendance management.

5. Examination and result management.

6. Basic academic report generation.

7. Role-based access.

Advanced features can be added in later releases.

18. Product Roadmap

Phase 1 – Domain analysis and problem understanding.

Phase 2 – Design Thinking and prototype.

Phase 3 – Agile sprint planning and backlog.

Phase 4 – Core application development.

Phase 5 – Testing and DevOps automation.

Phase 6 – Infrastructure and deployment improvements.

Phase 7 – Monitoring, feedback, and continuous improvement.

19. Initial System Architecture

Users → Web/Application Interface → Application/Business Logic → Database

Student, Faculty, Administrator and Examination Staff access the appropriate functions through the interface. The application layer processes registration, attendance, examination, result and reporting operations. The database stores student and academic records.

20. Project Repository Structure

```
student-record-management/
├── .github/            # CI/CD workflows
├── app/                # Application source code
│   ├── modules/        # Student, attendance, exam, result modules
│   ├── database/       # Database schemas/models
│   └── tests/          # Automated tests
├── docs/               # Project documentation
├── prototype/          # UI wireframes/prototype
├── deployment/         # Deployment configuration
└── README.md           # Project setup information
```

21. Project Charter

Project Name: Enterprise Student Record and Academic Management Platform

Project Type: Unit-1 Mini Project

Primary Goal: Design a centralized academic record-management product using Design Thinking.

Users: Students, faculty members, administrators, examination staff and IT support.

Main Deliverables: BRD, Product Vision, Stakeholder Analysis, User Personas, FRS, NFRS, Design Thinking Report, IBM Design Thinking Report, IBM Loop, MVP, Roadmap, Architecture, Repository Structure.

Success Criteria: Clear requirements, user-centred design, usable prototype, defined MVP, and a development-ready project plan.

22. Conclusion

The Student Record and Academic Management Platform provides a structured solution for managing academic information. The project applies the five Design Thinking stages, IBM Design Thinking principles, stakeholder analysis, user personas, requirements engineering, product vision, MVP planning, roadmap planning, and initial architecture. The result is a well-planned software product ready for Agile planning and later development.

23. Stakeholder Matrix

Stakeholder	Role	Expectation
Students	Use academic information	Easy access to records, attendance and results
Faculty	Manage academic activities	Simple attendance and record entry
Administrators	Monitor academic records	Centralized dashboard and reports
Exam Cell	Manage examinations/results	Accurate examination and result records
IT Support	Technical support	Secure and maintainable platform

24. MVP Feature Matrix

Feature	Priority	Release
Student Registration	High	MVP
Student Profile	High	MVP
Attendance	High	MVP
Examination & Results	High	MVP
Academic Reports	High	MVP
Advanced Analytics	Low	Future